Protocol for Treating Wound Healing (W.H) Using Low-Intensity Shockwaves Device:
Device: MoreNova DT, MoreNova Pro with C30 Transducer.
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1. Purpose and Scope

This protocol provides a standardized approach for wound healing such as- diabetic foot ulcers (DFUs), Venous Leg Ulcers and pressure wounds with Low-Intensity Extracorporeal Shockwave Therapy (LiESWT). It is intended for healthcare providers using the MoreNova Device with the C30 Transducer to promote wound healing, tissue regeneration and angiogenesis.

2. Introduction

- Venous Leg Ulcers: Caused by poor vein function that leads to fluid buildup and increased pressure, weakening the skin and slowing repair.
- Diabetic Foot Ulcers: Result from impaired blood flow and nerve damage, making injuries go unnoticed and healing inefficient.
- Pressure Sores: Appear when prolonged pressure compromises blood supply to the skin and underlying tissues, causing tissue breakdown and delayed recovery.

LiESWT has been shown to enhance blood flow, promote fibroblast proliferation, and accelerate tissue repair, making it a valuable adjunct to conventional wound care.

3. Background Condition Overview:

The wounds fail to heal naturally due to disrupted blood supply and persistent inflammation.

Instead of progressing towards recovery, the tissue remains stuck in a breakdown cycle, unable to transition into the rebuilding phase.

Mechanism of Action:

- Stimulation of Angiogenesis: LiESWT triggers the release of growth factors (such as VEGF and eNOS), which induce the formation of new blood vessels, restoring essential blood flow to the ischemic tissue.
- Tissue Regeneration: The therapy mechanically stimulates fibroblasts to increase collagen synthesis, which is critical for strengthening the tissue and accelerating wound closure.
- Anti-Inflammatory & Oxygenation: By enhancing local perfusion, LiESWT improves oxygen saturation and modulates inflammation, creating an optimal physiological environment for restarting the healing mechanism.

4. Patient Selection Criteria

Inclusion Criteria:

1. Adults (30–80 years) with chronic, non-healing DFUs (>4 weeks).
2. Patients with venous leg ulcers or pressure sores.
3. Post-surgical wounds with delayed healing.
4. Diabetic patients with adequate glycemic control (HbA1c < 8%).
5. ABI > 0.6 (to ensure adequate arterial supply).

Exclusion Criteria:

Electronic Devices & Metallic Implants

- Radiation-sensitive devices: Pacemakers, Cardioverters, Defibrillators (ICD), and Neurostimulators.
- Metallic or electronic implants: Any implant in or near the treatment area (e.g., metallic orthopedic screws, radioisotope needles).

Oncology & Radiotherapy

- Malignancy: Known or suspected malignancy at or near the treatment site.
- Cancer history: Patients recovering from cancer within the last 5 years.
- Past radiotherapy: Previous radiation treatment in the region being treated.

Hematological & Vascular Conditions

- Bleeding disorders: Clinically significant chronic hematological diseases.
- Blood thinners: Patients using anticoagulants (e.g., Coumadin) with an International Normalized Ratio (INR) > 3.
- Active bleeding: Presence of active bleeding or exposed major blood vessels.
- Ischemia: Critical Limb Ischemia (CLI) without restored perfusion or cases with insufficient arterial inflow.

Infections & Tissue Integrity

- Active infection: Local or systemic infections, including cellulitis, abscess, osteomyelitis, or sepsis.
- Necrotic tissue: Gangrenous tissue requiring surgical management or extremely fragile/necrotic tissue in the treatment area.
- Gas-filled organs: Treatment directly over organs such as the lungs or bowel.

Metabolic & Psychological Factors

- Metabolic control: Severely poor metabolic control (e.g., high HbA1c or uncontrolled hyperglycemia/diabetes).
- Psychological pathology: Psychiatric or psychological conditions that would interfere with or complicate the treatment.

5. Pre-Treatment Screening

To determine the patient's suitability for LiESWT, conduct the following:

- Comprehensive medical history and wound assessment, including size, depth, and condition of the wound.
- Documentation of wound status with photographs and measurements.
- Neuropathy testing using monofilament or vibration tests.
- Diagnostic imaging: X-ray or MRI to rule out underlying conditions like osteomyelitis.
- Vascular assessment: Doppler ultrasound to evaluate blood flow and arterial/venous supply.

6. Treatment Protocol Preparation:

- Cleanse the wound and surrounding skin.
- Apply a sterile, water-based coupling gel on and around the wound area.

Procedure:

- **Number of Sessions:** 6–12 sessions depending on the wound severity.
- **Session Frequency:** Twice per week.
- **Session Duration:** 7–20 minutes.
- **Applications Per Session:** Position the transducer **above and around** the wound, ensuring full coverage of affected tissue.
- **Shocks Per Application:** 500-750
- **Shocks Per Session:** depending on wound size.

Treatment Parameters:

- **Energy (HV) Settings:** 1.9 kV.
- **Frequency (FRQ):** 2 Hz.

7. Treatment Areas and Techniques

- Target the wound edges, and surrounding tissue to stimulate blood flow and angiogenesis.
- Avoid direct application over exposed bone or tendon.
- For larger wounds, treat in sections to ensure full coverage.

8. Patient Comfort and Safety

To ensure patient comfort and safety:

- Use sterile coupling gel to prevent infection and enhance shockwave transmission.
- Monitor patient feedback and adjust settings if discomfort occurs.
- Discontinue treatment if excessive pain or adverse reactions are noted.

9. Post-Treatment Care

- Reapply sterile dressings after treatment.
- Advise patients to avoid excessive movement or strain on the treated area.
- Maintain blood sugar control and foot hygiene.
- Use offloading techniques (e.g., special footwear) to prevent pressure buildup on the ulcer.

10. Treatment Outcomes and Follow-Up

Expected Results:

- Visible improvement in wound healing within 3–6 weeks.
- Reduction in wound size and improved tissue quality.
- Reduced risk of amputation in chronic diabetic wounds.

Follow-Up Visits:

- Regular assessments every 2 weeks to document progress.
- Adjust treatment parameters based on wound healing rate.
- Consider maintenance therapy every 3–6 months for chronic cases or slow-healing wounds.

11. Quality Control Device Calibration:

- Verify device settings before each session.

The MoreNova Device ensures preprogrammed parameters are applied accurately.

- **Clinician Training:**

- Operators must be trained in the use of the MoreNova Device and C30 Transducer.
- Review the “Instructions for Use” manual before initiating treatment.

- **Checklist for Each Session:**

- Confirm patient suitability and review contraindications.
- Ensure the transducer and device are clean, sanitized, and functioning correctly.
- Document session details, including energy settings, transducer placement, and patient response.

12. References

For clinical studies and additional resources, visit the Initia medical Clinical Data Center.